What teachers need to know about prematurity

Preterm birth

A full term pregnancy lasts approximately 37 to 40 weeks of gestation. A premature or preterm birth is where a baby is born before 37 weeks of gestation. Medical advances over the last 30 years mean that more preterm babies survive than ever before.

How common is this condition?

In the UK, around 8% of all babies are born preterm. This equates to approximately 60,000 preterm births each year and means that, on average, two or three children in every primary school class are likely to have been born preterm.

The more preterm a baby is born, the more medical care they need around the time of birth. Most preterm babies are born just a few weeks early, at 32 to 36 weeks of gestation, although some are born and survive as early as 22 weeks. Around 2% of all babies are born very preterm, before 32 weeks of gestation. Very preterm babies are very vulnerable at birth and may spend many weeks or months being cared for in a specialist neonatal unit before being discharged home.

What is the prognosis/long term view?

Research has shown that babies born preterm are more likely to have developmental or learning difficulties than babies born at term. Specifically, children born preterm are more likely to have difficulties with attention, problem solving, working memory, processing speed and hand-eye coordination than their term-born peers. They are also more likely to be shy or withdrawn, have difficulties interacting with peers, and have emotional problems such as anxiety, worries and fears (Wolke et al., 2019).

A small proportion of children born preterm are at increased risk for neurodevelopmental disorders, including autism spectrum disorder (ASD) and attention-deficit/hyperactivity disorder (ADHD). Even if they do not reach the criteria for a diagnosis they may display many behaviours associated with these conditions.

Children born preterm are also more likely to have poorer motor development than children born at term and to have poorer manual dexterity, balance and coordination. In some cases children may have suffered brain injury leading to cerebral palsy, but this is uncommon even after extremely preterm birth (Wolke et al., 2019). Preterm babies are at slightly higher risk of being deaf than babies born at term, although deafness affects only 1-2% of very preterm babies. A very small proportion of children born preterm may develop a condition called chronic lung disease of prematurity, which is not unlike asthma and may lead to breathing difficulties later in life.

Preterm birth can affect development throughout childhood and into adulthood. As a group, preterm children tend not to catch up with their peers. The latest research shows that, on average, preterm born children remain at higher risk for cognitive and social-emotional problems as adults and have lower paid jobs than adults born at term (Wolke et al., 2019).

How variable are the symptoms and experiences of this condition for young people?

The more preterm a baby is born, the more likely they are to have developmental or learning difficulties later in life. This means that extremely preterm babies, born before 28 weeks of gestation, are most likely to have difficulties in the long term.

However, outcomes following preterm birth are highly variable. It is important to recognise that not all children born preterm will have developmental or learning difficulties. Prematurity is just a risk factor for long term problems, it is not a diagnosis. Some babies born extremely preterm, at 23 or 24 weeks of gestation, may go on to have no difficulties at all, whereas other babies born just a few weeks early might have significant difficulties that impact on their learning and performance at school. Although most children born preterm won't have severe disabilities, many might have subtle cognitive, motor, social or emotional difficulties that impact on their learning.

How is this condition likely to impact on my student's learning?

The kinds of difficulties described above can have a major impact on a child's learning and performance at school. As a group, students born preterm are at higher risk for special educational needs than their term-born peers. For example, approximately 1 in 20 children born at term have special educational needs compared with 1 in 3 children born extremely preterm (before 28 weeks of gestation; Mackay et al., 2010). Compared with their term-born classmates, students born preterm have also been shown to have poorer attainment across all school subjects, but are especially likely to have difficulties with mathematics. A study in the UK has shown that 34% of children born very preterm had difficulties learning mathematics in Key Stage 2 (7-11 years), compared with just 16% of their term-born classmates (Simms et al., 2015).

How is this condition likely to impact on my student's behaviour?

Children born very preterm, before 32 weeks of gestation, are three times more likely to have ADHD than children born at term (Wolke et al., 2019). ADHD is diagnosed when children have persistent difficulties with attention or hyperactivity and impulsivity. Children with ADHD who were born preterm primarily have problems with attention; difficulties with hyperactivity or impulsivity are less common. This is an important difference between preterm children and other children with ADHD. As children born preterm with attention problems tend not to be hyperactive or disruptive, their difficulties can be harder to spot in the classroom and they may be overlooked (Brogan et al., 2014).

How is this condition likely to impact on my student's wellbeing?

Children born preterm often direct their feelings and emotions inward. They can appear withdrawn, shy and awkward; clingy in new situations; nervous, anxious, worried or fearful; or feel sad, lonely or depressed. This group of emotional problems is known as *internalising problems.* Signs of internalising problems may also include frequently complaining of illnesses such as headaches, stomach aches or sickness. Children born preterm are more likely to have these internalising problems than children born at term (Wolke et al., 2019).

How is this condition likely to impact on my student's interactions with others?

Children born preterm are more likely to have difficulties with social skills than children born at term. These include difficulties interacting with other children, making and maintaining friendships, working together in groups and communicating with peers.

Children with poor social skills may also be shy or withdrawn in social situations and might prefer the company of adults over children. Children with attention problems often find it harder to make friends and interact with other children. These kinds of difficulties may lead to children becoming isolated at school. Research has also shown that children born very preterm are more likely to be bullied than other children. Being bullied and having problems making friends can have a negative impact on a child's health and well-being and can lead to mental health problems later in life (Wolke et al., 2015).

How is all of this likely to impact on my student's family and what might be their greatest fear or challenge?

The experience of preterm birth and having a baby cared for on a specialist neonatal care unit can be distressing for parents and families. For many parents neonatal care is just the start of their journey as their children may go on to have developmental or learning difficulties as they grow up.

Some parents prefer not to disclose their child's birth history to the school, whilst others are in favour of the school knowing this information. Parents can sometimes find it challenging to convince teachers that their child's prematurity may affect their development or learning and might need additional support in school. It is important that teachers are aware of the kinds of difficulties preterm born children might face so they can provide them with appropriate support at the earliest opportunity and help allay parents' worries and concerns.

What can teachers do to help?

Teachers can help by increasing their knowledge of the long term impact of preterm birth to help them identify children with difficulties and provide appropriate support. To support children's learning and development, teachers can use evidence-based strategies that are likely to be beneficial for any child with the kinds of difficulties most commonly associated with preterm birth.

As noted above, children born preterm tend to struggle with attention, working memory, processing speed and hand-eye coordination. Strategies that could help include breaking instructions and assignments into chunks and encouraging students to focus on one task at a time, perhaps also providing written instructions or memory aids. Children with these difficulties may also need extra time to complete a task, as well as regular prompts or checks to keep them focused. When you ask a question it can help to wait a few moments before eliciting a response so that children with slower processing speed have the chance to answer. Writing guides and well-spaced worksheets can help children with poor hand-eye coordination.

If you notice that a child has difficulties characteristic of being born preterm, you may want to consider asking parents about their child's birth history. Some schools now routinely ask this when children start school. Regardless of whether you know a child's birth history, the advice and strategies described above are likely to be beneficial to any child with difficulties

in these areas. Try to identify each child's unique strengths and needs to guide selecting the most appropriate strategies to use with each individual child.

What other professionals might be involved?

As a group, children born preterm are more likely to access support from a range of health and education professionals, including speech and language therapists, occupational therapists, physiotherapists and clinical or educational psychologists.

As outcomes following preterm birth are highly variable, an individual assessment is needed to identify each child's unique support needs.

Is there a specialist charity or organisation to draw on for support or resources?

Yes. The PRISM Study Team led by the University of Leicester have developed an e-learning resource to help educational professionals improve their knowledge of the long term consequences of preterm birth and how to support preterm born children at school. The resource was co-designed with teachers, educational psychologists, parents of children born preterm and young adults who were born preterm themselves. It includes the latest evidence about what preterm birth is, how it may affect children's development and learning, and what can be done to support these children in the classroom.

The resource is freely available to use online at www.pretermbirth.info thanks to funding from children's charity Action Medical Research. Research has shown that use of the e-learning resource significantly improves teachers' knowledge of preterm birth and their confidence in supporting preterm born children in the classroom (Johnson et al., 2019).

There are a number of charities that specialise in supporting parents of children born preterm, including The Smallest Things (www.thesmallestthings.org), BLISS (www.bliss.org.uk) and the Irish Neonatal Health Alliance (www.inha.ie).

From October 2020, UK charity The Smallest Things are inviting schools to become 'Prem Aware' and achieve The Smallest Things' Prem Aware Award. Further details about the scheme and how your school can achieve the award are available online at www.thesmallestthings.org/prem-aware-award.

References

Brogan E, Cragg L, Gilmore C, Marlow N, Simms V & Johnson S. Inattention in very preterm children: implications for screening and detection. *Archives of Disease in Childhood* 2014; 99(9), 834-839.

Johnson S, Bamber D, Bountziouka V, Clayton S, Cragg L, Gilmore C, Griffiths R, Marlow N & Wharrad HJ. (2019). Improving developmental and educational support for children born preterm: evaluation of an e-learning resource for education professionals. *BMJ Open* 2019; 9(6):e029720.

MacKay DF, Smith GC, Dobbie R, Pell JP. 2010. Gestational age at delivery and special educational need: retrospective cohort study of 407,503 schoolchildren. *PLOS Medicine* 2010; 7:e1000289.

Simms V, Gilmore C, Cragg L, Clayton S., Marlow N & Johnson S. Nature and origins of mathematics difficulties in very preterm children: a different etiology than developmental dyscalculia. *Pediatric Research* 2015; 77(2):389-395.

Wolke D, Johnson S, Mendonca M. The Life Course Consequences of Very Preterm Birth. *Annual Review of Developmental Psychology* 2019;1:69–92.

Wolke D, Baumann N, Strauss V, Johnson S, Marlow N. Bullying of preterm children and emotional problems at school age: Cross-culturally invariant effects. *Journal of Pediatrics* 2015; 166:1417-22.

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